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Stratified assembly pressing for enhanced mechanical characteristics and sealing performance of borosilicate glass composites

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Abstract

Borosilicate glass composites are extensively used in glass-to-metal seals (GTMs) due to their superior insulating properties and hermeticity. In this study, the conventional dry-pressing process was optimized into a stratified assembly pressing method, where samples were compacted in 1, 3, or 5 layers and then pressed as an assembly to produce borosilicate glass composites with varying alumina doping concentrations. Experimental results show that increasing the number of stratified layers effectively reduces internal porosity. Furthermore, the "low porosity layer" structure formed during assembly pressing significantly decreases the free energy at crack tips, thereby enhancing the mechanical properties of the composites. The highest compressive strength, 658±24MPa, was achieved at an alumina doping level of 10wt%. However, excessive alumina doping resulted in particle agglomeration and increased porosity, which negated the benefits of the stratified assembly pressing and led to a reduction in compressive strength. To better evaluate sealing performance, the GTM shear strength testing method was improved by replacing the ISO 3611:2010 cross-cut method with a ring-seal configuration. This new configuration demonstrated that stratified assembly pressing enhances the compressive stress exerted by the metal shell on the borosilicate glass composite by modulating glass shrinkage during cooling. The optimal shear strength, 152±4MPa**,** was also obtained at 10wt% alumina doping, further confirming that the intrinsic properties of the glass play a pivotal role in seal performance. This study presents a novel and practical strategy for improving the quality and reliability of GTMs by optimizing both fabrication processes and testing methodologies.

Previous

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Keywords

Glass-to-metal seals; Stratified assembly pressing; Borosilicate glass composites; Mechanical properties; Testing methodology

1. Introduction

Glass-to-metal sealing (GTMs) is widely utilized in the nuclear power, electronic packaging, and new energy sectors due to its excellent insulation, aging resistance, and sealing reliability [[1], [2], [3]]. However, unlike metals, which possess superior mechanical properties, glass, being inherently brittle, constitutes the weakest component in the mechanical performance of GTMs, thereby limiting the broader application and development of glass materials [4]. This limitation stems from the intrinsic brittleness of glass, which exhibits a linear stress-strain response and lacks plastic deformation, often resulting in catastrophic failure under tensile stress.

To overcome the inherent brittleness of glass, researchers have extensively investigated doping strategies aimed at enhancing its mechanical properties. These approaches typically involve the incorporation of high-performance materials, such as particulate fillers, fibers, or woven fiber networks, into the glass matrix through physical bonding mechanisms. The resulting reinforcement mechanisms, including crack deflection, crack bending, crack bridging, and crack trapping, collectively reduce stress concentrations and enhance fracture resistance [[5], [6], [7]]. In addition to these physical effects, chemical interactions induced by dopants also play a critical role in performance improvement. For example, nucleating agents can promote crystallization, leading to composite microstructures comprising crystalline or amorphous phases, while fluxing agents modify the properties of the glass melt, facilitating densification and phase transitions during processing [8,9]. These strategies enable the design of tailored microstructures that balance mechanical robustness with functional performance, offering promising pathways for the advancement of glass materials in demanding applications.

While complex doping strategies are commonly employed to enhance material properties, optimizing the manufacturing process of glass presents a cost-effective and straightforward alternative for performance improvement. In the field of ceramics, studies have shown that adjusting compaction pressure and dwell time during the dry-pressing of powder precursors can significantly enhance densification through the plastic deformation of individual particles. This mechanism markedly improves the sintered density of the final product, thereby increasing both strength and toughness [10,11]. By designing tailored pressing protocols for glass powders, it is anticipated that their mechanical properties can be systematically and effectively optimized.

Previous studies have explored layered design concepts in metallic materials using techniques such as magnetron sputtering, powder metallurgy, and physical/chemical vapor deposition. These methods enable spatial variation of material properties through alternating distributions along the thickness direction, resulting in enhanced performance characteristics [[12], [13], [14]]. Layered structural designs have shown great promise in materials science due to their superior mechanical properties. Similarly, research on functionally graded materials (FGMs), particularly in ceramic or ceramic-metal composites, utilizes concentration and structural gradients to improve high-temperature resistance and mechanical performance [15,16]. Inspired by these advancements, the present study introduces a stratified pressing process for glass, aimed at mitigating its inherent brittleness and expanding the applicability of GTMs. By optimizing the compaction protocol, this approach seeks to develop tailored microstructural gradients that balance mechanical integrity with functional reliability, thereby addressing the limitations of conventional glass formulations in demanding operational environments.

2. Materials and methods

2.1. Sample preparation

The borosilicate glass powder used in this study, referred to as TH-3 glass, was prepared at the Beijing Key Laboratory of Fine Ceramics. Alumina particles (RUISiL, China) with an average particle size of 1 μm were incorporated into the glass powder at mass percentages of 0 wt%, 5 wt%, 10 wt%, 15 wt%, and 20 wt% as experimental variables. The powders were mixed with a polyvinyl alcohol (PVA) binder and granulated for the subsequent press-forming process. The mass of the powders was measured using a precision balance with an accuracy of 0.0001 g to ensure experimental consistency. Based on the increasing alumina doping content, the experimental samples were designated into five groups, labeled A (0 wt%) through E (20 wt%).

The dry-pressing process for the glass composite preforms involved two stages: stratified pressing and assembly pressing as shown in Fig. 1(a)–(c). In the stratified pressing stage, different amounts of powder were compacted layer by layer. This was followed by assembly pressing, in which the stratified samples were assembled and pressed again to form the final preforms. The variable mass of the powders allocated among different stratified assembly pressing serves solely to yield green compacts that conform to the test geometry uniformity. The number of layers used for assembly pressing included: 1 layer (1.20 g/layer), 3 layers (1.26 g/layer), and 5 layers (1.33 g/layer). The final height of the preforms showed no significant variation across the different layering configurations in Fig. 1(d). Stainless steel molds with a cross-sectional diameter of 5 mm were used for dry-pressing. To avoid the risk of delamination, during stratified pressing, the applied pressure was 0.8 ton with a dwell time of 1 min. In the subsequent assembly pressing stage, a pressure of 1.0 ton and a dwell time of 2 min were applied. Under consistent processing conditions, the powder mass for stratified assembly pressing was adjusted to produce cylindrical specimens with varying alumina doping contents, intended for measuring the coefficient of thermal expansion (CTE). The total layered masses for the samples were as follows: 1 layer (6.91 g), 3 layers (7.25 g), and 5 layers (7.67 g).

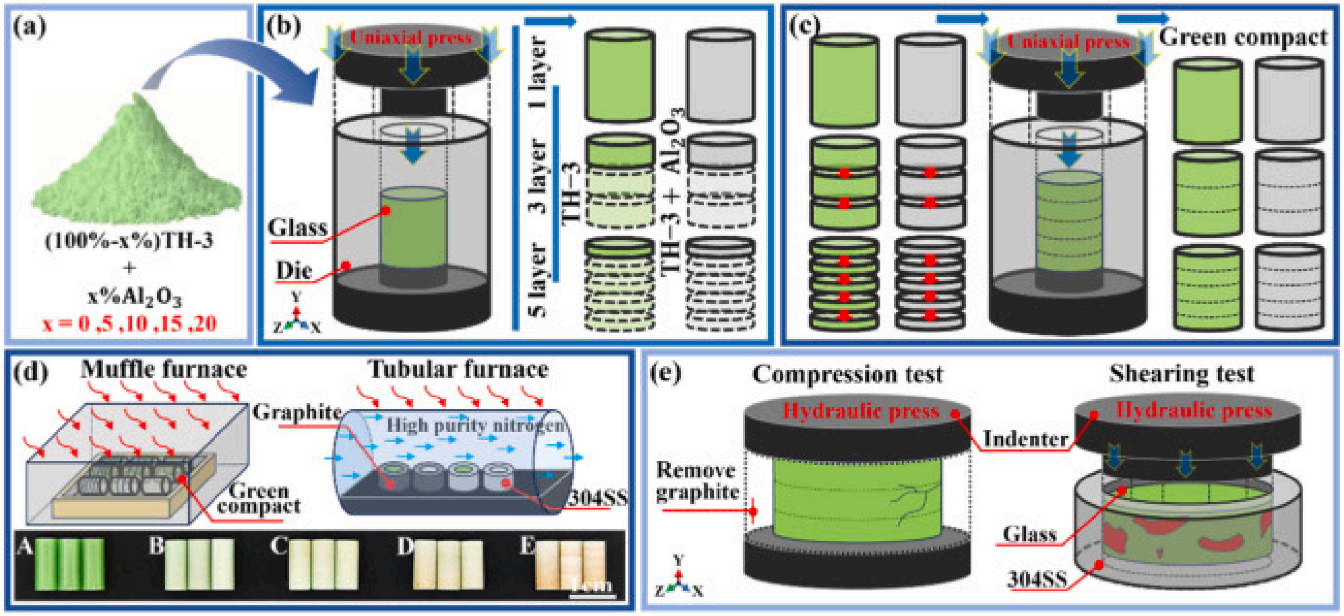

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Fig. 1. Manufacturing and testing processes of stratified assembly pressing.

The obtained glass preforms were initially sintered in a quartz furnace (SX2-5-12, CHOY, Shanghai, China) under an air atmosphere at 680°C for 1h to remove organic binders. Subsequently, the samples were transferred to a tube furnace (OTF-1200X, HF-Kejing, China) for further sintering and annealing under a nitrogen atmosphere, with a controlled gas flow rate of 1000mL/min at 1060°C [1,17]. The sintering protocol has been extensively detailed in previous studies. During the tube furnace sintering process, the vitrified samples were sealed with graphite and 304 stainless steel (304SS) shells (outer diameter: 10mm; inner diameter: 5mm; height: 15mm) to prepare specimens for compression and shear testing. To prevent dimensional changes caused by high-temperature wetting deformation, the upper surface of the glass samples was fixed during the sealing process. The detailed experimental procedures will be described in subsequent sections.

2.2. Experimental methods

The density of the experimental samples was measured using the Archimedes drainage method (Micromeritics, AccuPyc II 1345, USA). Internal porosity distribution was observed using an optical microscope (Carl Zeiss, Axio Scope A1, Germany), and porosity statistics were analyzed with Image-Pro Plus 6.0 software. A universal testing machine (AI-7000-MU, GOTECH, China) was used to perform compression tests on cylindrical specimens (diameter: 5 mm; height: 12.5 mm) to assess mechanical property variations and evaluate the shear strength of the GTMs.

To ensure the accuracy of compression test measurements, the specimens in Fig. 1(d) were polished using a metallographic grinder with 2000-grit sandpaper to achieve uniform dimensions (diameter: 5 mm; length: 12.5 mm). Scanning electron microscopy (SEM, SU8220, Hitachi, Japan) was employed to examine the fracture morphologies of both compression and shear specimens. Elemental distribution was analyzed using energy-dispersive X-ray spectroscopy (EDS, QUANTAX EDS XFlash®6, Bruker, Germany) with an acceleration voltage of 10 kV. Nano-indentation (Agilent Nano Indenter G200, USA) with Berkovich indenter (diamond three-sided pyramid, elastic modulus 1141 GPa, Poisson's ratio 0.07, 20 nm tip radius) was employed to characterize the mechanical properties of the borosilicate glass. A peak load of 10 mN was imposed and held for 10 s, with loading and unloading each performed over 40 s. The Young's modulus and hardness of the glass were extracted using the classical Oliver–Pharr model, whose detailed formulae have been given in our previous study [18]. To enhance surface conductivity for microscopic analysis, a thin platinum (Pt) film was deposited on the sample surfaces prior to imaging. The linear CTE was determined using a mechanical thermal analyzer (DIL 402 Expedis Supreme, Netzsch, Germany) under an inert argon atmosphere (flow rate: 0.04 L/min). The measurements were conducted with a heating rate of 5 °C/min over a temperature range of 25-600 °C.

Traditional testing of GTMs strength is based on the ISO 3611:2010 standard for evaluating the shear bond strength of fine ceramic interfaces [17], which employs a cross-cut method, as illustrated in Fig. 2(a). However, in GTMs, the bonding between glass and metal is not solely due to elemental diffusion, it also involves compressive stress exerted by the metallic shell [18]. The ISO 3611:2010 standard does not adequately replicate the compressive bonding characteristics of GTMs, primarily due to the high length-to-width ratio of the metal rod, which leads to non-uniform circumferential shrinkage in the glass during cooling. To address this limitation, a novel shear strength testing method, depicted in Fig. 2(b). This optimized approach effectively simulates the metal-induced compressive stress in GTMs by ensuring uniform circumferential pressure, while also meeting the requirements for sealing integrity assessment. By bridging the gap between conventional laboratory testing and real-world application performance, this method offers a more reliable and practical framework for evaluating the mechanical and sealing properties of GTMs composites under demanding service conditions.

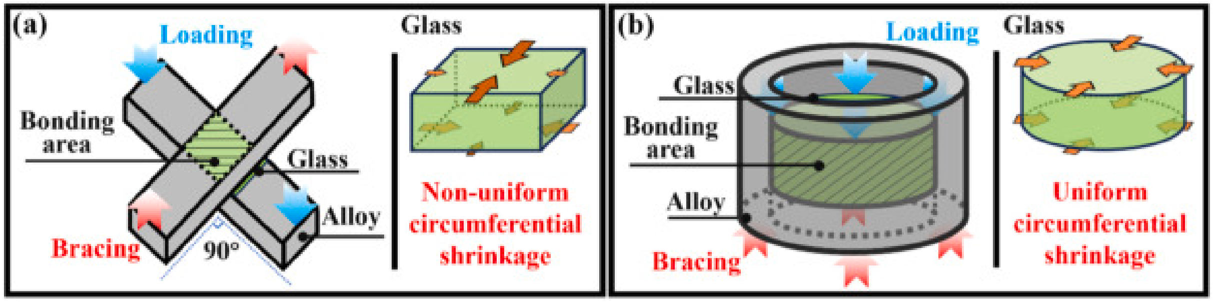

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Fig. 2. Comparison of bonding shear test methods: (a) ISO 3611:2010 cross-cut method, and (b) Ring-seal method.

To ensure a uniform contact area between the glass and the metal shell during sealing, a graphite mold was used to constrain the height of the glass throughout the sealing process. Specifically, graphite punches were employed to confine the glass powder between the inner surfaces of the graphite or 304 stainless steel annular shells. This setup allowed gravity to guide the glass into forming a localized bond within the predefined space, as illustrated in Fig. 3(a). Subsequent screening of the experimental samples confirmed sufficient wetting and verified that the bonding area conformed to the sidewall dimensions shown in Fig. 3(b). Finally, room-temperature shear tests were performed using a cemented carbide indenter, as depicted in Fig. 3(c).

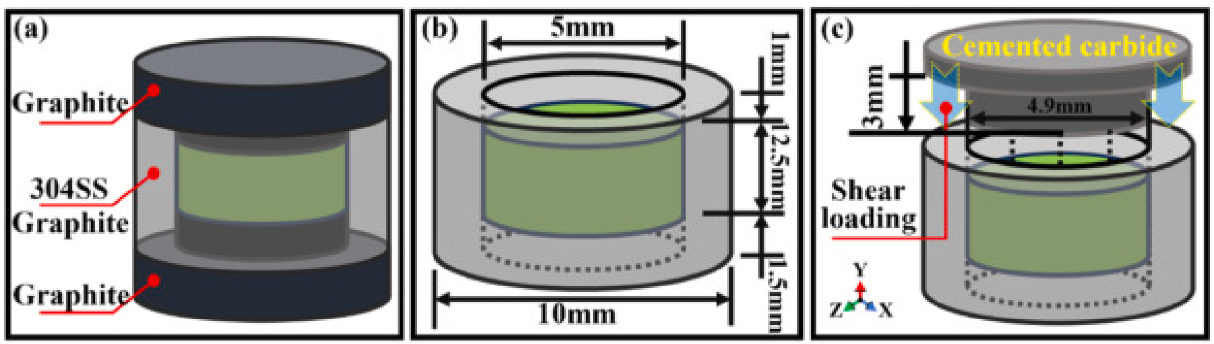

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Fig. 3. Ring-seal shear test: (a) Sample preparation, (b) Sample dimensions and (c) Shear method.

For every alumina doping glass subjected to stratified assembly pressing, five identical samples were prepared for compression testing, thereby ensuring the reliability of the experimental data. To minimize errors caused by wear of the loading platens and surface irregularities of the samples, an annealed copper foil (0.1 mm thick) was placed between the glass and the loading platens during both compression and shear tests, as reported in Ref. [19]. The loading rate was consistently maintained at 0.5 mm/min. The shear strength was calculated using the empirical formula (1), where τ represents the bonding shear strength, P is the maximum fracture load, and A is the effective bonding area. All experimental data are reported in accordance with the International System of Units (SI).

$$\tau = {}^{P}/_{A} \tag{1}$$

3. Results and discussions

3.1. Sample density

The relationship between the density of the experimental samples and the stratified assembly pressing process is summarized in Table 1. Although the mass percentage of alumina doped to the glass powder varied across samples, the density exhibited a consistent and predictable trend. Specifically, the sample density increased with higher alumina content; however, the rate of increase significantly declined beyond 10 wt%. This trend can be attributed to the higher density of alumina particles compared to the glass matrix, which initially contributed to overall densification. Nonetheless, the substantial hardness mismatch between alumina and glass particles led to preferential plastic deformation of the softer glass phase during compaction [20]. When the alumina content became excessive, particle agglomeration occurred, hindering the effective expulsion of interstitial gases from the powder beds and thus restricting further densification [21]. As a result, although the density generally increased with alumina content, the incremental gains progressively diminished due to the agglomeration-induced limitations.

Table 1. Sample density and group designations.

	Chemical composition	**1 layer**	**3 layers**	**5 layers**
Experimental grouping	TH-3 + 0 wt% alumina	2.4808 (A1)	2.4971 (A3)	2.5020 (A5)
	TH-3+5 wt% alumina	2.5236 (B1)	2.5345 (B3)	2.5484 (B5)

	Chemical composition	1 layer	3 layers	5 layers
	TH-3 +10 wt% alumina	2.5719 (C1)	2.5895 (C3)	2.5946 (C5)
	TH-3 +15 wt% alumina	2.5895 (D1)	2.5912 (D3)	2.6011 (D5)
	TH-3+20 wt% alumina	2.5907 (E1)	2.6063 (E3)	2.6108 (E5)

For experimental groups with identical alumina doping content, the sample density exhibited a positive correlation with the number of layers used in the assembly pressing process. This effect is primarily attributed to friction between the powder particles and the die during pressing, which causes a gradual axial reduction in the pressure transmitted by the die plunger. As a result, interstitial gases are not effectively expelled, leading to residual porosity after sintering and a consequent reduction in sample density [22]. Additionally, the friction between the die wall and powder particles influences the plastic deformation behavior of the powder bed. This leads to more pronounced deformation in the central region of the sample, as indicated by the red dashed line in Fig. 4(a)–(b). The uniaxial pressing method used in this study inherently produced a gradient density distribution, with lower density and greater heterogeneity observed in regions farther from the die plunger. Friction-induced pressure gradients during pressing have a substantial impact on densification mechanisms, particularly in dry-pressing processes. Under identical pressure conditions, increasing the number of stratified pressing layers reduces the powder mass per layer, facilitating more uniform compaction. This stratified configuration helps distribute the applied load more evenly across the powder bed, thereby enhancing the expulsion of trapped gas and increasing the green compact density.

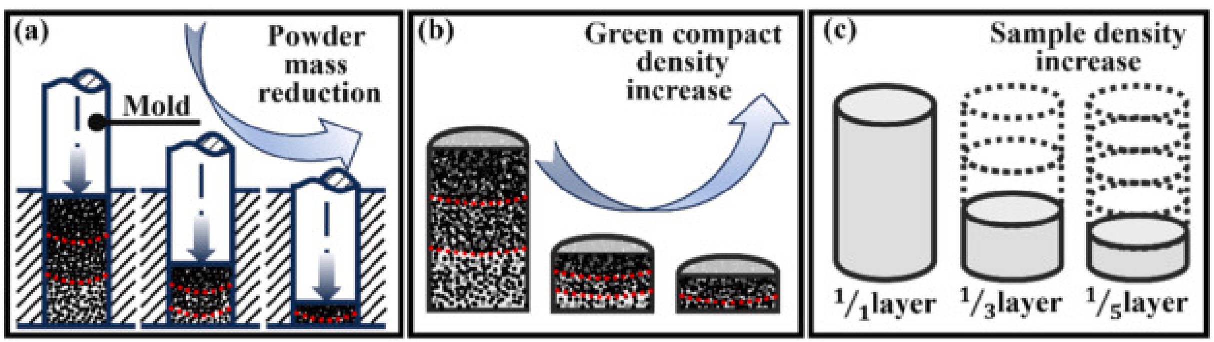

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Fig. 4. Density variation in stratified assembly pressing samples: (a)(b) Stratified pressing, and(c) Assembled pressing.

In the assembly pressing of samples, an increased number of layers corresponds to a higher degree of densification within each individual layer, thereby effectively enhancing the final density of the assembly pressed specimens. As shown in Fig. 4(c), the sample density increases significantly when the number of assembly layers reaches five. To promote further plastic deformation of the powder between layers during assembly pressing, the applied pressure and dwell time must be slightly higher than those used in the initial stratified pressing stage. Although friction between the powder and the die remains a contributing factor during assembly pressing, the pre-compacted density achieved during stratified pressing exerts a more significant influence on the final sample density than the frictional effects. This observation is consistent with previous findings, which emphasize that pre-compaction stages play a critical role in the overall densification process, particularly in layered manufacturing techniques [21].

The non-uniformity inherent in the dry-pressing process can affect the amount of residual air trapped between glass powder particles, thereby influencing the porosity of the sintered samples. To evaluate this effect, the sintered samples were polished to a mirror finish and examined under an optical microscope to observe porosity in the central region, as shown in Fig. 5. The final polishing site was fixed at half the sample height to provide an indirect estimate of the average pore population across the entire volume. For experimental groups with identical alumina doping content, increasing the number of layers in stratified assembly pressing significantly reduced both the overall porosity and the size of pores, confirming the effectiveness of this method in minimizing porosity, detailed data on the average porosity distribution of the samples are provided in the Supplementary Information Fig. S1. Notably, the C5 sample exhibited a porosity as low as 0.40%, substantially lower than that of other samples. However, as alumina doping content increased, the porosity of the sintered samples first decreased and then increased again. While stratified assembly pressing effectively reduced porosity in samples with 15 wt% and 20 wt% alumina particles, excessive doping resulted in a noticeable increase in porosity. Although alumina particles possess a higher mass density than borosilicate glass and generally contribute to improved sample density, excessive alumina doping led to particle agglomeration. This agglomeration hindered the plastic deformation of glass particles and restricted the viscous flow of molten glass during sintering [23]. This phenomenon is further supported by the observed change in porosity morphology, from the regular, circular pores in Fig. 5(a) to the irregularly shaped pores in Fig. 5(e), which aligns with the trends seen in the density variation of the samples.

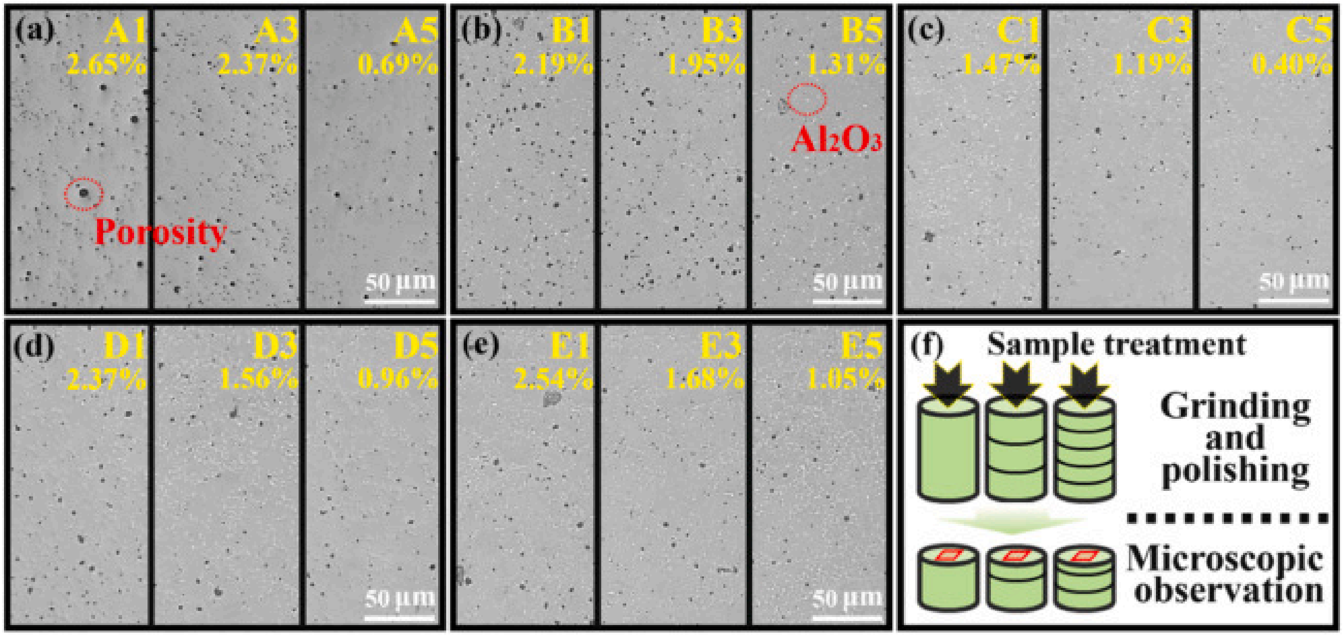

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Fig. 5. Average porosity of stratified assembly pressed samples: (a-e) 0 wt%, 5 wt%, 10 wt%, 15 wt%, 20 wt% alumina doping, and(f) Sampling location.

3.2. Compressive strength

Compression tests were conducted at room temperature on samples with varying alumina doping contents. Load-displacement curves were recorded for selected specimens in Fig. 6. The results indicate that, for specimens with identical alumina content, the compressive strength increased significantly with the number of assembly layers. This improvement is primarily attributed to the higher porosity typically observed in non-stratified pressing specimens. During sintering, the solidification and cooling shrinkage of these pores contribute to stress concentrations within the glass matrix. When propagating cracks encounter such porosities, they may be deflected or pinned due to localized stress field disturbances, thereby consuming additional fracture energy and enhancing the apparent toughness of the material [24]. However, when crack lengths are much larger than the pore diameter, substantial deflection is unlikely, and the presence of numerous pores becomes ineffective in hindering crack propagation, potentially accelerating failure within the specimen [25,26]. Therefore, in the compressive failure experiments of glass specimens, stratified assembly pressing reduces internal porosity, thereby effectively enhancing compressive strength.

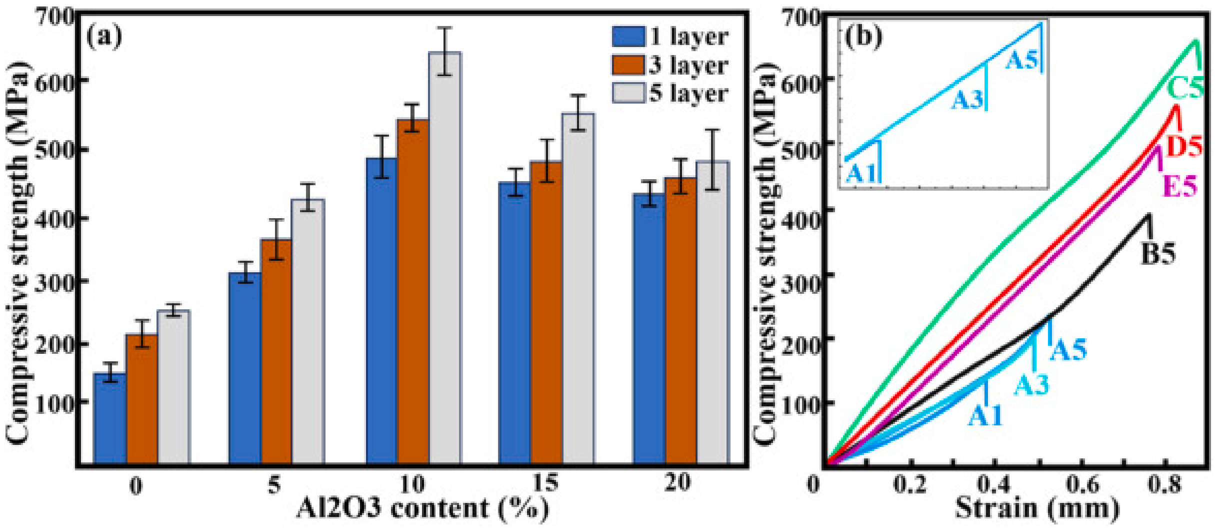

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Fig. 6. Compressive test results: (a) Compressive strength of the specimen, and (b) Partial measured values.

With increasing alumina doping content, the compressive strength of the experimental groups followed a trend of initially increasing and then decreasing. As shown in Fig. 6(a), the sample with 10wt% alumina doping achieved the highest compressive strength, reaching 658±24MPa through stratified assembly pressing. However, further increases in alumina content led to a gradual decline in compressive strength. This behavior can be attributed to the dual role of alumina in the glass matrix. At moderate doping levels, alumina effectively promotes crack deflection, which enhances the material resistance to fracture and improves its mechanical properties. In contrast, excessive alumina doping impairs the plastic deformation of glass particles and hinders the viscous flow of the sintered glass, primarily due to particle agglomeration. This agglomeration also obstructs the escape of internal gases during sintering, contributing to increased porosity [23]. Although alumina particles can reduce the free energy at crack tips and serve as effective toughening agents, the presence of numerous entrapped pores counteracts these benefits, significantly weakening the reinforcing and toughening effects of the particles.

In addition, compared to the stratified pressing samples, the assembly-pressed samples exhibited a distinct "low porosity layer" structure, as highlighted in the red area in Fig. 7, with an approximate thickness of 10 μm. Moreover, it was found that as the number of stratified assembly pressing layers increases, the porosity of the "low porosity layer" reaches a minimum at 10 wt% alumina doping. This phenomenon can be attributed to differences in the distribution of frictional forces during the pressing process. In stratified pressing, the plastic deformation of glass particles at the top and bottom surfaces of the powder compact is less affected by the sidewall friction of the mold, resulting in a more compacted surface. During assembly pressing, further plastic deformation occurs at the interfaces between layers, leading to the formation of a more densely packed "low porosity layer" at these boundaries. The presence of this structure plays a critical role in hindering internal crack propagation, thereby significantly enhancing the compressive strength of the samples. This observation further supports the conclusion that stratified assembly pressing is an effective method for improving the mechanical performance of glass-based composites.

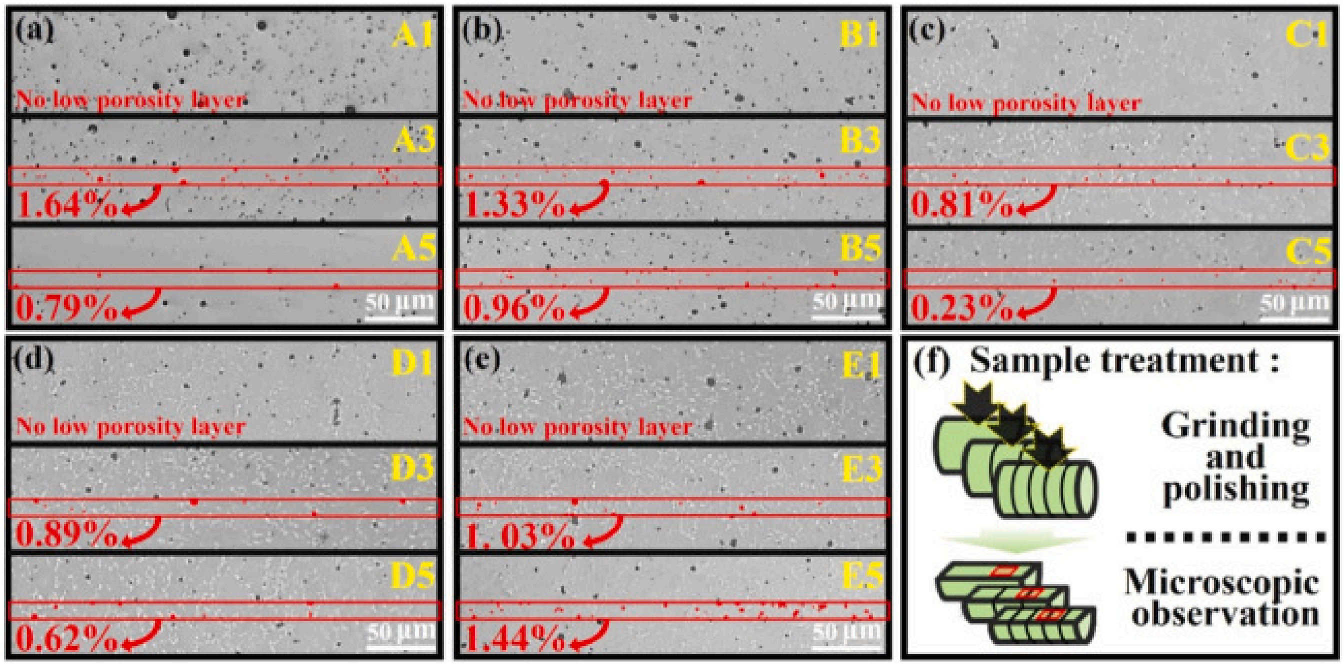

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Fig. 7. Porosity of stratified assembly pressing "low porosity layer" structure: (a-e) 0 wt%, 5 wt%, 10 wt%, 15 wt%, 20 wt% alumina doping, and (f) Sampling location.

The incorporation of alumina significantly enhances the compressive strength of borosilicate glass composites, as demonstrated by the marked improvement in their mechanical properties. However, when the alumina doping content exceeds 10 wt%, the characteristic “low porosity layer” structure gradually deteriorates. This degradation is primarily attributed to the increased presence of alumina particles within these regions. During reassembly pressing, the plastic deformation of the surrounding glass powder is impeded by the rigid alumina particles, leading to the formation of new porosities and consequently weakening the integrity of the “low porosity layer.” Detailed data on the porosity distribution of the low-porosity layer are provided in the Supplementary Information Fig. S2. Furthermore, prior studies have reported that high concentrations of reinforcement phases tend to induce particle agglomeration, which disrupts the viscous flow of glass during high-temperature processing. But no chemical interaction occurs between the alumina and the borosilicate glass matrix during high-temperature sintering. Alumina therefore functions only as physically toughening particles dispersed in the glass [26]. Such agglomeration exacerbates the formation of localized porosities around alumina clusters, thereby further diminishing the effectiveness of the “low porosity layer” formed between layers. These experimental findings are corroborated by SEM and EDS results presented in Fig. 8, where the irregular porosities observed in Fig. 8(c) are directly linked to excessive alumina content. Ultimately, this structural degradation contributes to the observed reduction in compressive strength of the doped composites.

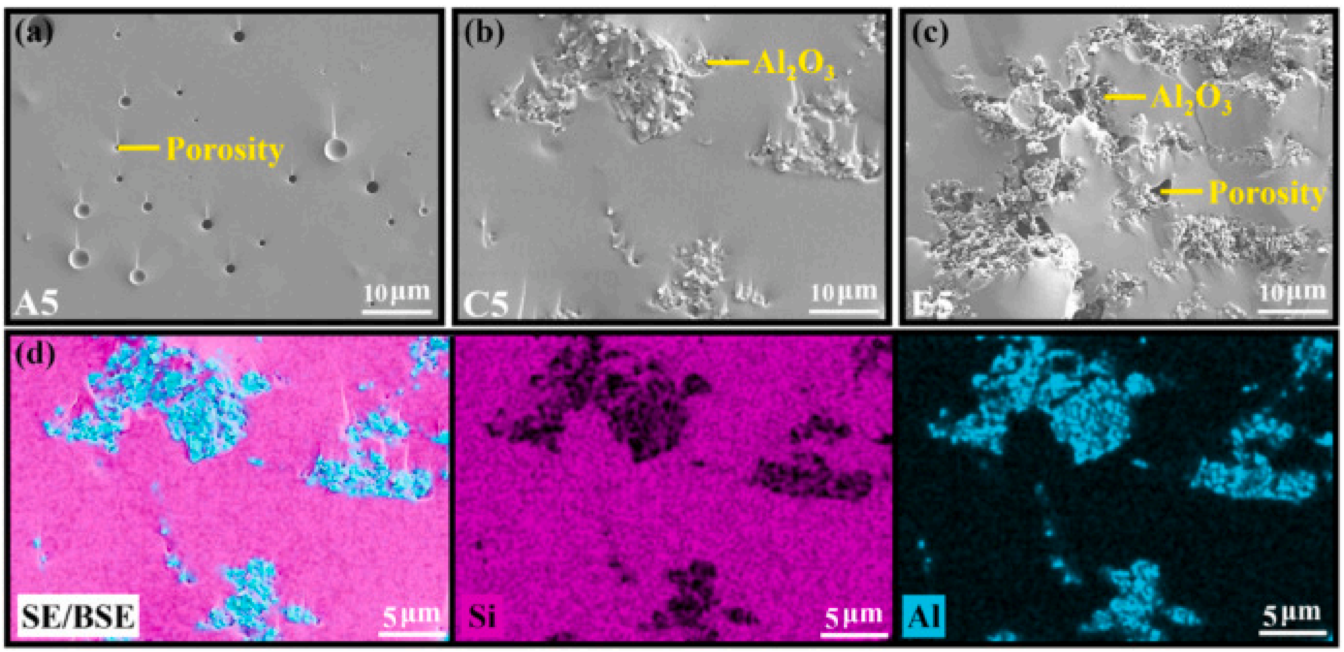

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Fig. 8. Effect of alumina doping level on porosity: (a-c) Group A5, C5, E5, and (d) Elemental distribution in Group C5.

The schematic diagram in Fig. 9 illustrates the plastic deformation mechanism of glass powder during the stratified assembly pressing process. As depicted in Fig. 9(a)–(c), pure glass powder undergoes plastic deformation under stratified pressing. Residual gas trapped within interparticle gaps often fails to escape during high-temperature processing, leading to the formation of porosity. The introduction of harder alumina particles intensifies the plastic deformation of the surrounding glass powder, allowing it to conformally encapsulate the alumina particles. Nonetheless, excessive alumina doping leads to particle agglomeration, which significantly hinders the encapsulation process and obstructs the escape of residual gas during subsequent sintering. This results in the formation of gas pores around the alumina particles. These observations align with findings that particle agglomeration diminishes the effective interaction between the glass and alumina phases, thereby compromising the overall densification of the composite. Moreover, the influence of particle size distribution on gas permeation and interfacial bonding is supported by previous studies on glass-alumina composites [27].

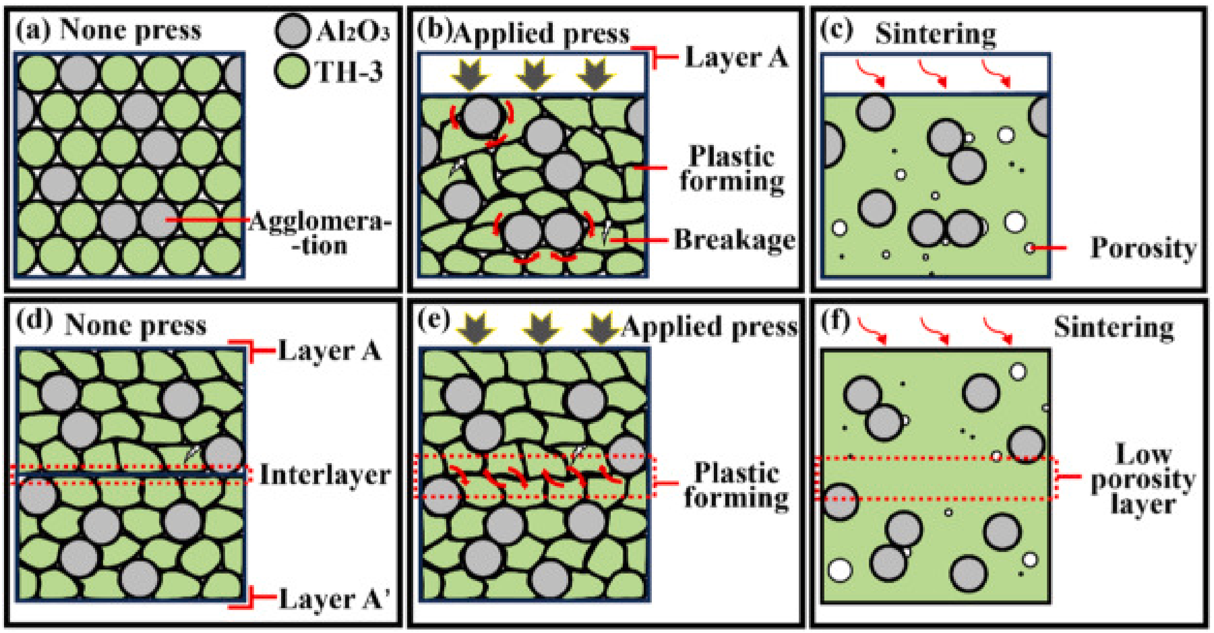

Fig. 9. Formation mechanism of internal porosity and the "low porosity Layer" structure in specimens: (a-c) Stratified pressing plastic deformation, and (d-f) Assembly pressing plastic deformation.

Fig. 9(d)–(f) illustrate the assembly pressing process, in which stratified pressed green bodies (Layer A and Layer A′) are re-pressed under mold pressure. The upper and lower circular cross-sections of these stratified pressed bodies, subjected to direct extrusion by the mold, exhibit higher density compared to the lateral surfaces influenced by mold friction. During assembly pressing, the glass particles undergo further plastic deformation, ensuring tight interlayer bonding and promoting densification of the upper and lower surfaces, as shown in Fig. 9(e). This results in the formation of the "low porosity layer" structure. However, excessive alumina doping inhibits plastic deformation at the upper and lower surfaces, while alumina agglomeration introduces additional trapped gas, significantly reducing the effectiveness of the "low porosity layer." This behavior aligns with the observation that particle agglomeration restricts gas escape during sintering, causing porosity formation around alumina particles.

For the A1 group shown in Fig. 10(a), simply increasing the dry-pressing pressure and dwell time proved insufficient to effectively reduce the internal porosity of the samples. As a result, irregular pores persisted within the sintered specimens, creating localized stress concentrations during cooling [28]. Porosity promotes microcrack propagation, significantly compromising mechanical strength. In contrast, the A5 group depicted in Fig. 10(b), fabricated via stratified assembly pressing, exhibited substantially reduced porosity size and a distinct "low porosity layer" structure. Under compressive loading, microcrack propagation triggered by pores was effectively suppressed, while the dense "low porosity layer" further impeded crack extension, thereby significantly enhancing compressive strength. The reduced residual crack length observed at the pore trailing edges on the fracture surface of the A5 group sample in Fig. 10(b) further validates this mechanism.

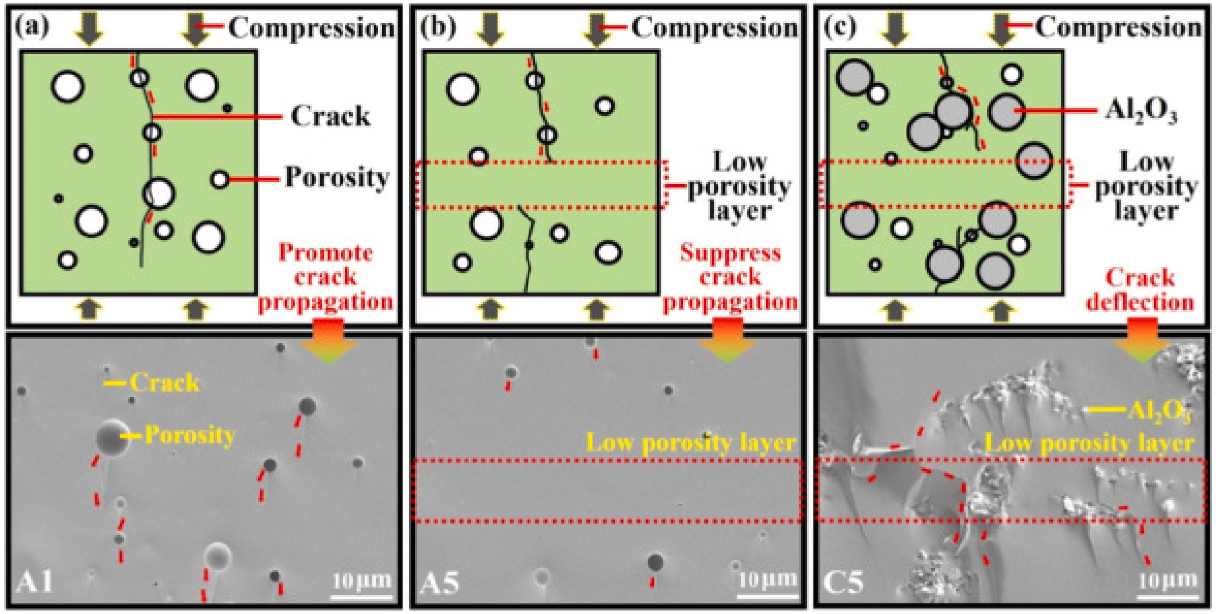

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Fig. 10. Fracture morphology and failure mechanism of specimens under compression: (a) Group A1, (b) Group A5 and (c) Group C5.

Existing studies have demonstrated the effectiveness of particle reinforcement in toughening glass. The reinforcement phase consumes fracture energy through mechanisms such as crack deflection and interfacial propagation, thereby enhancing the material compressive strength [29,30]. The schematic diagram of compression shown in Fig. 10(c) effectively illustrates the toughening mechanism induced by alumina doping, while the formation of a “low porosity layer”, further facilitates pronounced crack deflection at the fracture surface, contributing to improved compressive strength of the borosilicate glass composites. However, excessive alumina doping leads to a significant increase in porosity, which adversely affects the mechanical properties of the samples. This observation aligns with previous findings indicating that excessive reinforcement phases may cause particle agglomeration, hinder the densification process, and ultimately reduce the effectiveness of the reinforcement mechanism [26].

In addition, to further validate the effectiveness of the stratified assembly pressing in enhancing the mechanical performance of borosilicate glass, nano-indentation was performed to verify both the presence and the significance of the “low-porosity layer”.

To preclude inadvertent omission of the “low porosity layer” near the surface, an array of 3 ∗ 6 indents (spacing 5 μm) was executed at predefined locations. The data confirm the existence of this “low porosity layer” produced by the stratified assembly pressing in Fig. 11. Its hardness and reduced modulus substantially exceed those of the surrounding regions, corroborating the efficacy of the process in enhancing the mechanical performance of borosilicate-glass-matrix composites. Moreover, statistical scatter in the nano-indentation results increases monotonically with alumina content, being most pronounced at 20 wt %, indirectly demonstrating that excessive alumina doping limits further mechanical improvement.

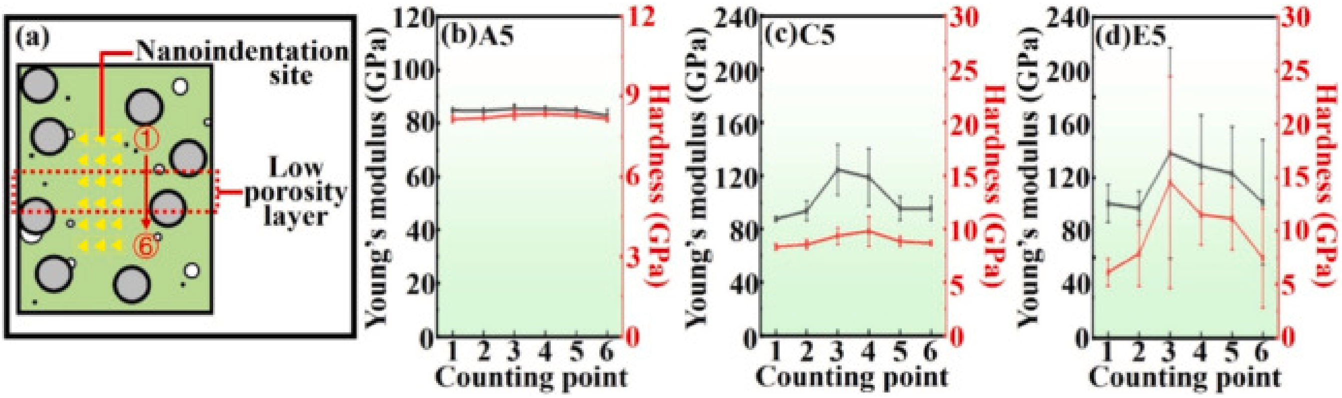

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Fig. 11. Nanoindentation tests: (a) Data acquisition locations, and (b-d) Data of group A5, C5 and E5.

3.3. Shear strength

To validate the feasibility of stratified assembly pressing for practical applications of glass matrix composites, shear and sealing tests were conducted, with the results presented in Fig. 12(a). Shear strength obtained by the conventional cross-cut method is approximately 37±3 MPa. In contrast, the newly proposed test method achieved a more than threefold increase in shear strength, fully demonstrating the feasibility of the new experimental approach. Moreover, the data indicate that the shear strength of the samples increases significantly as the number of layers in the stratified assembly pressing process rises. Fig. 12(b) summarizes the measured values for several experimental groups. This improvement is primarily attributed to the substantial reduction in internal porosity achieved with increasing layer count, which effectively mitigates interface debonding or delamination caused by boundary porosities [31,32].

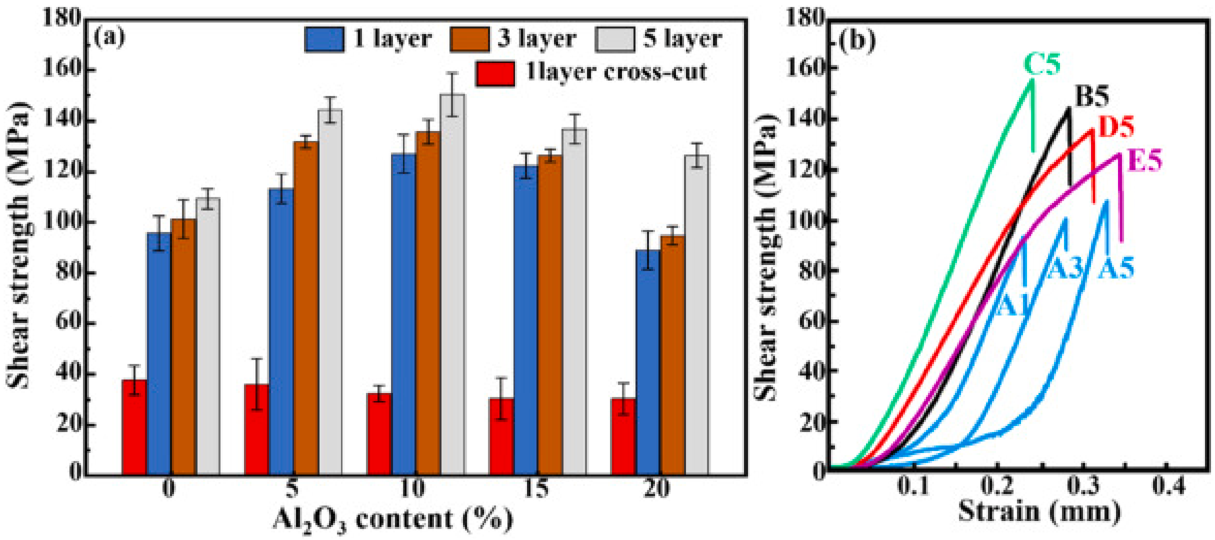

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Fig. 12. Shear strength test results: (a) Comparison of shear strength, and (b) Partial measured values.

Furthermore, the formation of a “low porosity layer” in the glass samples fabricated via stratified assembly pressing inhibits crack initiation within the glass matrix, redirecting crack propagation toward the GTMs interface. This mechanism contributes to the enhanced shear strength of the GTMs specimens. Regarding alumina doping content, the shear strength of the GTMs exhibits a pronounced dependence on doping level. The maximum shear strength of 152±4MPa was attained at 10wt% alumina doping under the stratified assembly pressing process. However, further increases in alumina content resulted in a marked decline in shear strength. This trend aligns with previous findings that excessive reinforcement phases can induce particle agglomeration and impede densification, ultimately diminishing the efficacy of the reinforcement mechanism. It should be noted that reliable GTMs relies on a synergy of mechanical compression and chemical bonding. Alumina doping does not alter the chemical bonding, which indicates that no chemical interaction was detected within the resolution limits of the employed characterization techniques. Instead, it influences the physical bonding by modifying the CTE of the material [1].

To elucidate the influence of porosity and alumina doping content on the shear strength of GTMs, the CTE of various glass composites was measured, with the results presented in Fig. 13. The stratified assembly pressing process effectively reduces porosity while causing only a slight decrease in the CTE of the glass. In contrast, the incorporation of alumina markedly lowers the CTE, exhibiting a positive correlation with doping content. This indicates that the CTE of the glass is primarily governed by its chemical composition and microstructure rather than the quantity of internal porosities [[33], [34], [35]]. The shear strength at the GTMs interface depends chiefly on the chemical bonding at the interface and the compressive stress applied by the metal shell. During the cooling process following sealing, the contraction of the glass corresponds to its thermal expansion behavior, implying that a lower CTE results in reduced volumetric shrinkage. A lower CTE (or reduced shrinkage volume) of the glass generates an increased compressive stress within the metal shell. Consequently, a stronger compressive force from the metal shell favors improved adhesion between the glass and metal. These findings are consistent with previous studies demonstrating that the mechanical behavior of GTMs is governed by interfacial compatibility and the residual stress distribution induced by thermal mismatch [36,37].

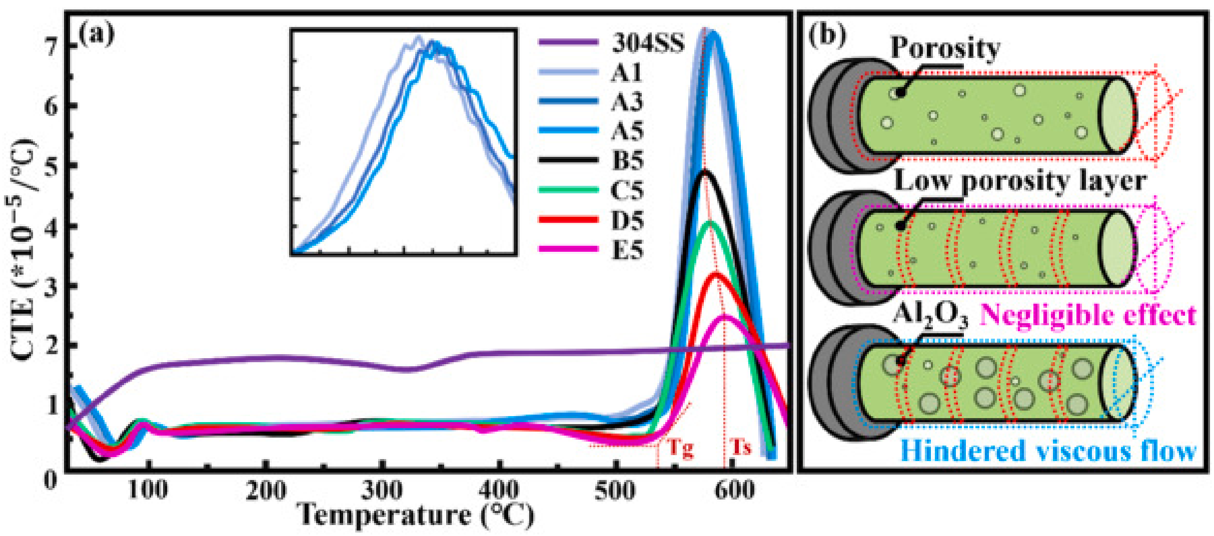

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Fig. 13. CTE of the test materials: (a) Measured values, and (b) Mechanism of thermal expansion mismatch.

Excessive alumina doping significantly reduces the shear strength of GTMs, primarily due to the obstruction of gas pore escape from the glass matrix and the substantial weakening of the “low porosity layer” structure formed during stratified assembly pressing. Although the CTE of the glass governs the magnitude of the compressive stress exerted by the metal shell, doping with alumina further reduces the glass CTE. However, the presence of numerous internal porosities creates stress concentration sites that promote crack propagation through the glass phase rather than confining cracks to the interface, thereby undermining the overall mechanical performance.

Meanwhile, the measurements shown in Fig. 13(a) reveal that the glass-transition temperature (*T*g) and the softening point (*T*s) of borosilicate glass increase monotonically with alumina content, consistent with earlier reports that additional alumina suppresses viscosity (η) [[38], [39], [40]]. Since the GTMs sealing temperature (1060 °C) greatly exceeds both *T*s and *T*g, the key issue is the influence of glass viscosity on shrinkage during cooling. As the alumina content increases, the progressive rise in viscosity becomes one of the dominant factors reducing the overall sealing contraction of the glass. Stratified assembly pressing reduces this shrinkage by lowering the internal porosity, whereas alumina doping further diminishes it by hindering viscous flow. Fig. 13(b) schematically illustrates how these two factors jointly modify the overall cooling contraction of the borosilicate glass.

Consequently, the schematic diagram in Fig. 14 further illustrates the differences in the cooling process during sealing. The stratified assembly pressing method reduces internal porosity in the glass, forming a “low porosity layer” structure that minimizes thermal contraction during cooling. Under identical deformation conditions, this enhances the compressive effect exerted by the metal shell on the glass, thereby significantly improving the shear strength of the GTMs interface, as shown in Fig. 14(a)–(b). For alumina doping borosilicate-glass composites, the lower CTE of alumina relative to that of the glass, combined with its suppression of viscous flow, effectively mitigates glass shrinkage during the sealing and cooling processes. This further strengthens the compressive force applied by the metal shell, as depicted in Fig. 14(c). However, excessive alumina doping fails to eliminate residual porosities caused by particle agglomeration, resulting in significant glass shrinkage during cooling. Consequently, the compressive effect of the metal shell is diminished, leading to a reduction in shear strength at the GTMs interface. Moreover, excess alumina hinders the viscous flow of glass, preventing gas pore escape and weakening the integrity of the “low porosity layer” structure. This deterioration compromises the mechanical properties of the glass, causing shear failure to initiate within the glass matrix rather than at the interface, further reducing the overall shear strength.

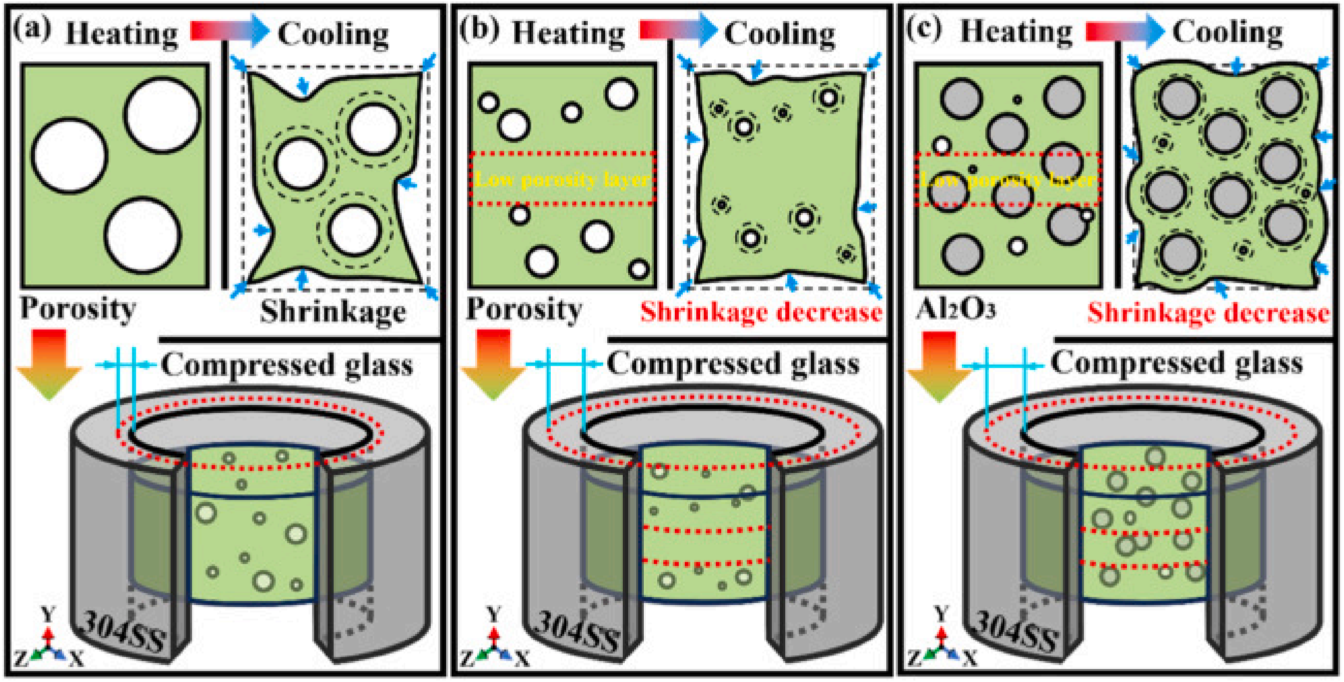

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Fig. 14. GTMs shrinkage mismatch: (a) Conventional dry-pressing, (b) Stratified assembly pressing, and (c) Stratified assembly pressing with alumina doping.

Different pressing techniques and alumina doping levels not only influence the shear strength of GTMs but also result in distinct variations in shear fracture morphology, as illustrated in Fig. 15. For borosilicate glass without alumina doping, the shear fracture after sealing primarily manifests as cracks at the interface, accompanied by arc-shaped cracks near this interface. As the number of layers in the stratified assembly pressing process increases, internal cracks within the glass progressively shift closer to the interface in Fig. 15(b). This behavior is attributed to the fixed bonding strength at the interface and the intrinsic strength of the glass. When the glass strength is relatively low, shear loading tends to initiate cracks at the weaker regions, causing cracks to preferentially nucleate within the glass matrix and propagate toward the interface. With an increasing number of pressing layers, the number of pores near the interface markedly decreases, while the formation of the “low porosity layer” enhances the glass resistance to cracking. This improvement promotes a gradual transfer of the shear fracture from the glass interior toward the interface. Such a mechanism effectively dissipates fracture energy at the crack tip and significantly improves the shear strength of the GTMs in Fig. 15(b)–(c).

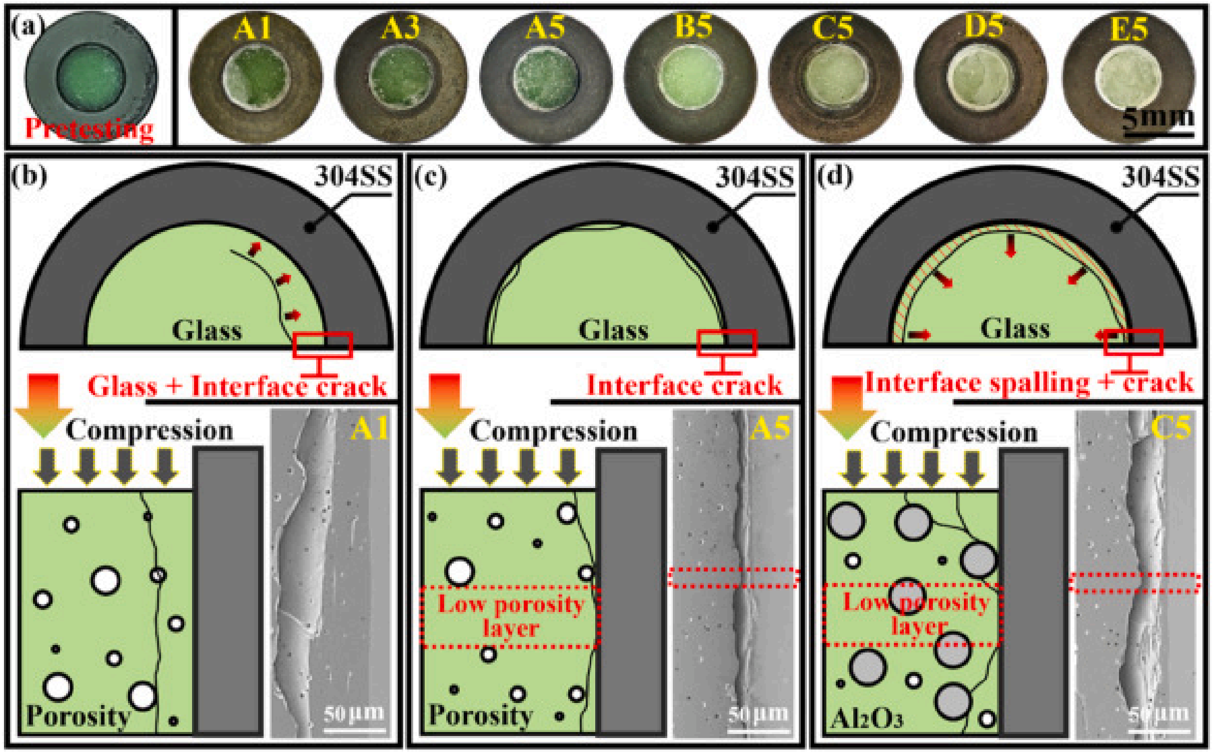

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Fig. 15. Shear Fracture: (a) Specimen morphology, and (b-d) Fracture mechanism of group A1, A5 and C5.

Appropriate alumina doping effectively enhances the crack resistance of the glass matrix. Under shear loading, fracture in GTMs primarily occurs at the interface, as demonstrated by the fracture morphology of sample B5 in Fig. 15(a). Alumina doping promotes crack deflection within the glass matrix, shifting fracture initiation closer to the interface and thereby improving the shear strength of the GTMs. However, this crack deflection mechanism also induces ring-like delamination near the interface, as observed in sample C5 in Fig. 15(a). With further increases in alumina doping content, excessive agglomeration of alumina particles leads to significant porosity within the glass matrix, which cannot be effectively eliminated by stratified assembly pressing. This deterioration compromises the mechanical properties of the glass, causing fracture initiation within the weaker glass phase under repeated shear loading. Consequently, the shear strength of the GTMs decreases, accompanied by an increase in surface delamination area due to compressive stress imposed by the metal shell, as evidenced by the partial D5 and complete E5 delamination shown in Fig. 15(a). Notably, despite these changes, the structural integrity of the GTMs remains intact before and after shear testing, with fracture predominantly localized near the interface, thus validating the efficacy of the experimental methodology.

4. Conclusions

In this study, the dry-pressing process was optimized by implementing a two-step procedure consisting of stratified pressing followed by assembled pressing. The compressive strength and shear strength of the GTMs interface were evaluated under this modified processing protocol, alongside the development of a novel ring-seal testing method. Through comprehensive analysis of porosity distribution, microstructural characteristics, and CTE measurements of the stratified assembly pressed specimens, the fundamental mechanisms driving the enhancement of mechanical properties were elucidated. The key findings are summarized as follows:

Stratified assembly pressing significantly reduces internal porosity in the pressed samples, with an increasing number of layers further decreasing porosity. However, excessive alumina doping beyond 10 wt% weakens this effect due to particle agglomeration. The optimal compressive strength of 658±24 MPa was achieved at 10 wt% alumina doping, confirming the effectiveness of the stratified assembly pressing structure in mitigating porosity.

Stratified assembly pressing enhances the shear strength of the GTMs interface, with strength increasing as the number of layers grows. Both porosity and alumina content influence the CTE, which in turn affects the compressive stress applied by the metal shell during cooling. Alumina doping primarily governs the CTE response, with 10 wt% alumina yielding the maximum shear strength of 152±4 MPa. Over-doping, however, reduces the glass mechanical strength, leading to interfacial failure within the glass phase and subsequent strength degradation.

The ring-seal testing method is better suited for evaluating GTMs shear strength under extreme conditions. Compared to the ISO 3611:2010 standard, it provides improved control over high-temperature wetting deformation and compressive stress in the metal shell, resulting in higher accuracy and a more realistic simulation of operational conditions.

CRediT authorship contribution statement

Chao Zhou: Writing – original draft, Validation, Formal analysis, Data curation, Conceptualization, Writing – review & editing. **Keqian Gong:** Validation, Investigation, Formal analysis, Data curation. **Zheng Liu:** Validation, Supervision, Resources, Investigation. **Zifeng Song:** Software, Resources, Methodology, Investigation. **Zhangjing Shi:** Visualization, Resources, Formal analysis. **Siyue Nie:** Validation, Investigation. **Yong Zhang:** Writing – review & editing, Resources, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

The following is the supplementary data to this article:

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Multimedia component 1.

Recommended articles

Data availability

Data used in this manuscript will be made available by the corresponding author upon reasonable request.

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